

Preferred Name

Surname (Family Name)

YEAR 13 MATHEMATICS

END OF TOPIC TEST 2 (Mechanics Side)

- Arithmetic Sequences
- Geometric Sequences
- Iterative Sequences

Duration: 40 minutes



For Teacher use only		
Question	Maximum Mark	Mark Awarded
1	6	
2	7	
3	7	
4	6	
5	6	
Total	32	

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen.

Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, ask for additional paper and write your name at the top.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Sufficient working must be shown to demonstrate the **mathematical** method employed.

Answers without working may not gain full credit.

Unless the degree of accuracy is stated in the question, answers should be rounded appropriately.

You are reminded of the necessity for good English and orderly presentation in your answers.

REFLECTION (to be completed once this test has been marked and returned to you)

Question 1

The fastest speed of the car

- Given that the fastest speed of the car in successive gears is modelled by an **arithmetic sequence**,

(3)

[illegible]

(b) find the fastest speed of the car in 5th gear.

(3)

[illegible]

[illegible]

(i) In an arithmetic series, the first term is a and the common difference is d .

$$S_n = \frac{n}{2} [2a + (n-1)d] \quad (3)$$
[illegible]

- Given that James takes n weeks to save exactly £64

- (b) Solve the equation

(c) Hence state the number of weeks James takes to save enough money to buy the printer, giving a brief reason for your answer.

(1)

This image shows a full page of a document template designed for handwritten notes or essays. It features approximately 28 evenly spaced, thin grey horizontal lines across the entire width of the page. The margins are consistent on all sides, providing ample space for writing. There are no pre-printed questions, headings, or other markings on the page.

[illegible]

[illegible]

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 5

In this question you should show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

A company made a profit of £20 000 in its first year of trading, Year 1

A model for future trading predicts that the yearly profit will increase by 8% each year, so that the yearly profits will form a geometric sequence.

According to the model,

(a) show that the profit for Year 3 will be £23 328

(1)

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(b) find the first year when the yearly profit will exceed £65 000

(3)

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Turn over.

(2)

This image shows a full page of a handwriting practice worksheet. It consists of approximately 20 horizontal rows. Each row is defined by two parallel dotted lines, creating a series of uniform gaps for letter height. The entire page is otherwise blank, with no margins, text, or other markings.

Reminder: Sufficient working must be shown to demonstrate the **mathematical** method employed.

Question 1

A car has six forward gears.

The fastest speed of the car

- in 1st gear is 28 km h^{-1}
- in 6th gear is 115 km h^{-1}

Given that the fastest speed of the car in successive gears is modelled by an **arithmetic sequence**,

(a) find the fastest speed of the car in 3rd gear.

(3)

Given that the fastest speed of the car in successive gears is modelled by a **geometric sequence**,

(b) find the fastest speed of the car in 5th gear.

(3)

5 (a)	Uses $115 = 28 + 5d \Rightarrow d = (17.4)$	M1	3.1b
	Uses $28 + 2 \times "17.4" = \dots$	M1	3.4
	$= 62.8 \text{ (km h}^{-1} \text{)}$	A1	1.1b
		(3)	
(b)	Uses $115 = 28r^5 \Rightarrow r = (1.3265)$	M1	3.1b
	Uses $28 \times "1.3265^4" = \dots$ or $\frac{115}{"1.3265"}$	M1	3.4
	$= 86.7 \text{ (km h}^{-1} \text{)}$	A1	1.1b
		(3)	

(6 marks)

Question 2

- (i) In an arithmetic series, the first term is a and the common difference is d .

Show that

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

(3)

- (ii) James saves money over a number of weeks to buy a printer that costs £64

He saves £10 in week 1, £9.20 in week 2, £8.40 in week 3 and so on, so that the weekly amounts he saves form an arithmetic sequence.

Given that James takes n weeks to save exactly £64

- (a) show that

$$n^2 - 26n + 160 = 0$$

(2)

- (b) Solve the equation

$$n^2 - 26n + 160 = 0$$

(1)

- (c) Hence state the number of weeks James takes to save enough money to buy the printer, giving a brief reason for your answer.

(1)

Turn over.

13 (i)	States that $S = a + (a + d) + \dots + (a + (n-1)d)$	B1	1.1a
	$S = a + (a + d) + (a + (n-1)d)$ $S = (a + (n-1)d) + (a + (n-2)d) + \dots + a$	M1	3.1a
	Reaches $2S = n \times (2a + (n-1)d)$ And so proves that $S = \frac{n}{2}[2a + (n-1)d]$ *	A1*	2.1
		(3)	
(ii)	(a) $S = 10 + 9.20 + 8.40 + \dots$		
	$64 = \frac{n}{2}(20 - 0.8(n-1))$ o.e	M1	3.1b
	$128 = 20n - 0.8n^2 + 0.8n$ $0.8n^2 - 20.8n + 128 = 0$ $n^2 - 26n + 160 = 0$ *	A1*	2.1
		(2)	
	(b) $n = 10, 16$	B1	1.1b
		(1)	
	(c) 10 weeks with a minimal correct reason. E.g. • He has saved up the amount by 10 weeks so he would not save for another 6 weeks • You would choose the smaller number • He starts saving negative amounts (in week 14) so 16 does not make sense	B1	2.3
		(1)	

(7 marks)

Question 3

The first three terms of a geometric sequence are

$$3k + 4 \quad 12 - 3k \quad k + 16$$

where k is a constant.

(a) Show that k satisfies the equation

$$3k^2 - 62k + 40 = 0 \quad (2)$$

Given that the sequence converges,

(b) (i) find the value of k , giving a reason for your answer,

(ii) find the value of S_{∞} (5)

9(a)	$\frac{12-3k}{3k+4} = \frac{k+16}{12-3k}$	M1	3.1a
	$3k^2 - 62k + 40 = 0 \quad *$	A1*	1.1b
		(2)	
(b)(i)	$3k^2 - 62k + 40 = 0 \Rightarrow k = \dots$	M1	1.1b
	States $k = 20$ and gives a reason e.g. that this gives a values of r such that $ r < 1$	A1	3.2a
(ii)	$a = 64$ and $r = -\frac{3}{4}$ (or allow $a = 6$ and $r = \frac{5}{3}$)	B1	1.1b
	$S_{\infty} = \frac{"64"}{1 - "(-\frac{3}{4})"} = \dots$	M1	3.1a
	$S_{\infty} = \frac{256}{7}$	A1	1.1b
		(5)	

(7 marks)

Turn over.

Question 4

A sequence $u_1, u_2, u_3 \dots$ is defined by

$$u_1 = 35$$

$$u_{n+1} = u_n + 7 \cos\left(\frac{n\pi}{2}\right) - 5(-1)^n$$

(a) (i) Show that $u_2 = 40$

(ii) Find the value of u_3 and the value of u_4

(3)

Given that the sequence is periodic with order 4

(b) (i) write down the value of u_5

(ii) find the value of $\sum_{r=1}^{25} u_r$

(3)

2(a)(i)	e.g. $(u_2 =) 35 + 7 \cos\left(\frac{\pi}{2}\right) - 5(-1)^1 = 40$ *	B1*	2.1
(ii)	$u_3 = 40 + 7 \cos\left(\frac{2\pi}{2}\right) - 5(-1)^2 (= 28)$ or $u_4 = "28" + 7 \cos\left(\frac{3\pi}{2}\right) - 5(-1)^3 (= 33)$	M1	1.1b
	$u_3 = 28$ and $u_4 = 33$	A1	1.1b
		(3)	
(b)(i)	$(u_5 =) 35$	B1	2.2a
(ii)	e.g. $\sum_{r=1}^{25} u_r = 6(35 + 40 + "28" + "33") + 35$	M1	3.1a
	$= 851$	A1	1.1b
		(3)	

(6 marks)

Question 5

In this question you should show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

A company made a profit of £20 000 in its first year of trading, Year 1

A model for future trading predicts that the yearly profit will increase by 8% each year, so that the yearly profits will form a geometric sequence.

According to the model,

(a) show that the profit for Year 3 will be £23 328 (1)

(b) find the first year when the yearly profit will exceed £65 000 (3)

(c) find the total profit for the first 20 years of trading, giving your answer to the nearest £1000 (2)

5(a)	$u_3 = £20\,000 \times 1.08^2 = (£) 23\,328^*$	B1*	1.1b
		(1)	
(b)	$20\,000 \times 1.08^{n-1} > 65\,000$	M1	1.1b
	$1.08^{n-1} > \frac{13}{4} \Rightarrow n-1 > \frac{\ln(3.25)}{\ln(1.08)}$ or e.g. $1.08^{n-1} > \frac{13}{4} \Rightarrow n-1 > \log_{1.08}\left(\frac{13}{4}\right)$	M1	3.1b
	Year 17	A1	3.2a
		(3)	
(c)	$S_{20} = \frac{20\,000(1-1.08^{20})}{1-1.08}$	M1	3.4
	Awrt (£) 915 000	A1	1.1b
		(2)	

(6 marks)

Turn over.